JULIA K. ELROD

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SUMMARY

I am a fifth-year Ph.D. student advised by Dr. Will Townes in the Department of Statistics & Data Science at Carnegie Mellon University (CMU). I am passionate about applying and developing statistical methods for genomics data. During my Ph.D., I have collaborated with the Torok lab at the University of Pittsburgh Medical Center (UPMC) studying pediatric scleroderma single-cell data. Before coming to CMU, I worked on cancer epidemiology at NIH. I am searching for a postdoctoral fellowship (to begin fall 2027) in statistical genomics. My long-term goal is to become faculty at an academic institution.

EDUCATION

Carnegie Mellon University, Pittsburgh, PA
Ph.D. Candidate in Statistics & Data Science Expected May 2027
Committee: Will Townes (Chair), Kathryn Roeder, Wei Chen, & Kathryn Torok.
Master of Science in Statistics May 2024

Kenyon College, Gambier, OH
Bachelor of Arts in Mathematics & Statistics, magna cum laude May 2020

RESEARCH EXPERIENCE

Carnegie Mellon University, Pittsburgh, PA Fall 2022–Present
Statistics Ph.D. Student | Advisor: Will Townes, Ph.D. (CMU)

- *Applied work: CITE-seq analysis of juvenile systemic sclerosis (jSSc) monocytes over time since autologous stem cell transplant (ASCT).*
  - Collaboration with Dr. Kathryn Torok's pediatric scleroderma lab at UPMC Children's Hospital.
  - *Goal:* Characterize cellular activity in jSSc patients before ASCT, over time since ASCT, and compared to healthy controls.
  - *Data preprocessing:* Implemented a comprehensive single-cell data processing pipeline in R, including quality control filtering, ambient RNA removal, doublet detection, and multimodal cell type annotation based on CITE-seq (RNA + protein) measurements.
  - *Longitudinal differential expression analysis:* Implemented a negative binomial generalized linear model regressing total gene expression per patient sample against time since ASCT.
- *Methodological work: Benchmarking single-cell RNA-seq normalization methods for accurate gene expression proportion recovery*
  - Evaluated novel and popular scRNA-seq normalization methods using various measures of distance (e.g., total variation distance, squared Hellinger distance, and Kullback–Leibler divergence) to compare ground truth and estimated expression distributions.

National Cancer Institute, National Institutes of Health (NIH), Bethesda, MD Fall 2020–Spring 2022
Postbaccalaureate CRTA Research Fellow | Supervisor: Philip Rosenberg, Ph.D.

- Developed novel statistical methodology in cancer surveillance.
- Created R software for the application of these methods by biomedical researchers.
- Methodology includes age-period-cohort modeling, kernel smoothing algorithms, model selection techniques, changepoint detection, and generalized linear models.
- Applications include quantification of trends in breast, pancreas, and oral cancer incidence, as well as clinical recommendations for patient hip fracture risk assessment.

The Ohio State University, Columbus, OH Summer 2018

Statistical Genetics Research Assistant | Supervisor: Asuman Turkmen, Ph.D.

- Compared the efficacy of three multivariate methods for detecting relationships between genes and disease risk in Genome-Wide Association Study (GWAS) data to three similar methods popular in the field of statistics.
- Conducted simulation studies and applied methods to Dallas Heart Study data, noting which methods picked up on the known relationship between rare variations in the ANGPTL5 gene and reduced triglyceride levels.

PUBLICATIONS

In Preparation

- **Elrod, J. K.** & Townes, F. W. (2026). Benchmarking single-cell RNA-seq normalization methods for accurate gene expression proportion recovery.

Preprints

- **Elrod, J. K.**, Sanyal, A., Hutchins, T., Townes, F. W., & Torok, K. S. (2026). Longitudinal single-cell modeling reveals monocyte reprogramming in juvenile systemic sclerosis following autologous stem cell transplantation. bioRxiv. https://doi.org/10.64898/2026.08.21.738279

Peer-reviewed

- Storandt, M. H., Tella, S. H., Wieczorek, M. A., Hodge, D., **Elrod, J. K.**, Rosenberg, P. S., Jin, Z., & Mahipal, A. (2024). Projected Incidence of Hepatobiliary Cancers and Trends Based on Age, Race, and Gender in the United States. *Cancers*, *16*(4), Article 684. https://doi.org/10.3390/cancers16040684
- Rosenberg, P. S., Filho, A. M., **Elrod, J.**, Arsham, A., Best, A. F., & Chernyavskiy, P. (2023). Smoothing Lexis diagrams using kernel functions: A contemporary approach. *Statistical Methods in Medical Research*, 09622802231192950. https://doi.org/10.1177/09622802231192950
- Zumsteg, Z. S., Luu, M., Rosenberg, P. S., **Elrod, J. K.**, Bray, F., Vaccarella, S., Gay, C., Lu, D. J., Chen, M. M., Chaturvedi, A. K., & Goodman, M. T. (2023). Global Epidemiologic Patterns of Oropharyngeal Cancer Incidence Trends. *JNCI: Journal of the National Cancer Institute*, djad169. https://doi.org/10.1093/jnci/djad169
- Allbritton-King, J. D., **Elrod, J. K.**, Rosenberg, P. S., & Bhattacharyya, T. (2022). Reverse engineering the FRAX algorithm: Clinical insights and systematic analysis of fracture risk. *Bone*, *159*, 116376. https://doi.org/10.1016/j.bone.2022.116376

TALKS

"Benchmarking single-cell RNA-seq normalization methods for accurate gene expression proportion recovery" October 2026

- Kathryn Roeder Lab (CMU, statistical genetics)

"Cross-tissue RNA-seq prediction with reduced rank regression" August 2026

- Joint Statistical Meetings (JSM) 2026, Boston, MA (accepted for oral presentation)

"Tips and tricks for a smooth workflow on the department Linux servers" April 2026

- CMU Statistics & Data Science Department (StatBytes seminar series)

"Longitudinal model of paired peripheral blood CITE-seq and skin scRNA-seq data in juvenile systemic sclerosis (jSSc) patients following autologous stem cell transplant (ASCT) reveals reduced expression of SSc marker genes" October 2025

- American College of Rheumatology (ACR) Convergence 2025, Chicago, IL (selected for oral abstract session)

"Longitudinal cross-tissue modeling of single-cell multi-omics data, with an application in juvenile systemic sclerosis" October 2025

- CMU Statistics & Data Science Department (thesis proposal)

"Longitudinal CITE-seq analysis of stem cell transplants in pediatric scleroderma"

- Paul Szabolcs Lab (UPMC, bone marrow transplantation and cellular therapies) June 2025
- Robert Lafyatis Lab (UPMC, systemic sclerosis) June 2025
- Kathryn Torok Lab (UPMC, pediatric scleroderma) April 2025, May 2025
- Kathryn Roeder Lab (CMU, statistical genetics) April 2025

"Evaluating genetic colocalization analysis techniques with an application in schizophrenia etiology" November 2023

- CMU Statistics & Data Science Department (Advanced Data Analysis project presentation)

"An R Package for Kernel Filtration of Rates on a Lexis Diagram" April 2021

- NIH Postbac Poster Day

"Using Multivariate Association Measures to Identify Relationships Among Genetic Variants and Multi-Dimensional Structured Traits" July 2018

- The Ohio State University Comprehensive Cancer Center & Kenyon College Pelotonia Partnership Undergraduate Research Symposium

TEACHING & MENTORING EXPERIENCE

Carnegie Mellon University, Pittsburgh, PA

Statistics Teaching Assistant

- Deep Learning (46-937) & Natural Language Processing (46-924) for M.S. in computational finance Fall 2025 & 2026
- Undergraduate Statistical Computing (36-350), Head TA Spring 2026
- Machine Learning II (46-927) for M.S. in computational finance Spring 2025
- Ph.D. Regression Analysis (36-707) Fall 2024
- Foundations of Data Science Online Graduate Certificate (36-640 and 36-641) Spring 2024
- Statistical Methods in Health Sciences (36-470) Spring 2024
- Undergraduate Advanced Data Analysis (36-402) Spring 2023
- Modern Regression (36-401) Fall 2022

National Institutes of Health, Bethesda, MD

College Summer Opportunities to Advance Research (CSOAR) Mentor Summer 2021

- Served as a primary mentor through CSOAR, which provides research opportunities to college students disadvantaged by circumstances that have negatively impacted their educational opportunities.
- Taught student age-period-cohort analysis and oversaw final project, "Recent Trends in Oral Tongue Cancer Incidence in the United States by Region."

Summer Internship Program (SIP) Mentor Summer 2021

- Co-mentored a college student through SIP, sharing the ins and outs of government biomedical research.
- Introduced student to statistical methodology in changepoint detection and complementary R software.
- Showed student how to run simulation studies comparing methods, resulting in the final presentation, "Estimating Changes in Average Annual Percentage Change of Disease Rates: Alternatives to Join-Point Regression."

Kenyon College, Gambier, OH

Career Services Associate Spring 2018–Spring 2020

- Workshopped résumés and cover letters with college students, helping students market their skills.

Apprentice Teacher of Spanish Fall 2017 & Spring 2020

- Taught supplementary Spanish lessons four times per week, with an emphasis on conversation practice.

Association for Women in Mathematics Secretary Spring 2018–Spring 2020

- Planned and organized mathematics education outreach events, such as a campus-wide STEM activities fair.

STEM and Data Teaching Assistant Summer 2017

- Collaborated with science faculty to introduce underrepresented first-year students to STEM at Kenyon.

AWARDS & CERTIFICATES

- Health Science Communication & Policy Workshop Certificate, NIH Office of AIDS Research Spring 2021
  - Studied the health policy legislative process, including securing research funding and resources.
- Fulbright English Teaching Assistant Award, Spain, 2020–2021 Spring 2020
  - Declined due to the COVID-19 pandemic.
- Sigma Xi – Scientific Research Honor Society, Kenyon-Denison Chapter Spring 2020
- Wendell D. Lindstrom Memorial Prize Spring 2018
  - Recognizes first- or second-year college students who have demonstrated great promise in mathematics.
- Pi Mu Epsilon – Honorary National Mathematics Society Spring 2018
- Kenyon College National Merit Scholarship Fall 2016–Spring 2020

MEMBERSHIPS

- American Statistical Association